Cybercrime

Lecture #15

Cybercrime

This is used to refer to three types of criminal activity

- Conventional crimes where cyberspace or computers are just an instrumentality – i.e. financial fraud
- The distribution of prohibited content – i.e. obscene material, hate speech
- Crimes against computers and computer infrastructure

A brief history of cybercrime

- 1970s – 1990s : The early years
  - Protest and concept demonstrations dominate
  - Culture of hacktivism
  - The US Computer Fraud and Abuse Act 1986 was one of the first legislations on the subject
  - The first viruses make their appearance
- The decade of the 2000s : The rise of organized cybercrime
  - A shift towards groups or organized criminals
  - Large attacks
  - The emergence of darknet marketplaces
  - A number of countries agree to cooperate in the area of cybercrime and The Budapest convention on Cybercrime comes into being

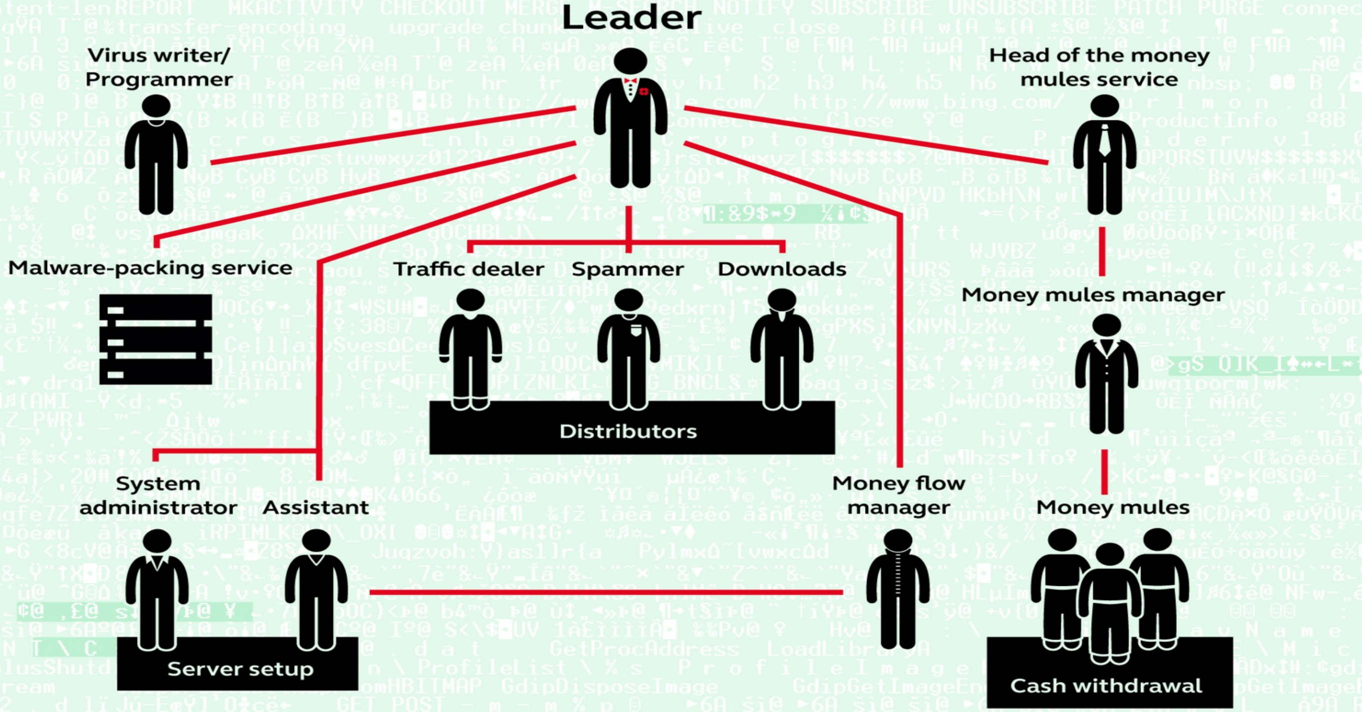
  Note: India is not a signatory to the Budapest convention

Cybercrime today

- Increased sophistication
- Cybercrime syndicates run as corporations
  - Have a well defined organization structure – CEO, CTO, Cyber operations, on ground operations
- Availability of many aspects of the criminal activity are available as services
  - Spam as a service
  - Hosting of phishing sites
  - Ransomware as a service
- Increased state sponsored activity
- Shift in victim profiles
- Greater collaboration amongst cybercriminals
- Evolution of a vast and diversified criminal and criminal support ecosystem
  - E.g. Jamtara, Mewat etc

How a financial cybercrime group is organized

Kaspersky Lab is actively investigating five large, Russian-speaking cybercriminal groups involved in stealing money using malicious software.



The Money flow manager transfers funds from attacked financial accounts to accounts provided by the Money mules manager. The Money mules manager instructs the money mules where to transfer the money. A share of the stolen money ends up with the Head of the money mules service, while the rest is transferred to the Leader of the criminal group.

Factors that greatly facilitate cybercrime

- Product developers are under severe financial pressure to produce new products and upgrade existing products. This means that functionality gets more attention than security
- A user of a modern computing device is just not able to comprehend how her device works and to determine if it might be compromised
- People have proven to be very susceptible to social engineering
- The widespread availability of electronic payment systems especially those that are cross-border and provide anonymity (such as bitcoin) greatly facilitate monetizing cybercrime and laundering the proceeds of crime in general

Jurisdictional challenges in addressing cybercrime

- In traditional crime the concept of jurisdiction is very important and well defined
- Since a cyber-transaction may involve people and ICT infrastructure in several different states, if this transaction is illegal then whose jurisdiction does this fall into?
- Jurisdiction is important for purposes of determining not only whose laws and regulations will apply but also who will carry out investigation, who will carry out prosecution, and who will carry out enforcement actions
- Cybercriminals understand this difficulty and carry out their activity in such a way that various jurisdictions are involved so as to hamper efforts to thwart their activity and bring them to book
- Cybercrime control therefore is essentially borderless and requires a great degree of collaboration between states and countries

The Budapest Convention

- The member states of the Council of Europe and a few other countries signed the Convention on Cybercrime (European Treaty Series 185) which was opened for signature on 23rd Nov 2001. This is popularly known as the Budapest Convention
- The objectives of this convention broadly are
  - To "harmonize" or create common definitions and understanding of various national laws on cybercrime
  - Helping states whose legislation and evidence frameworks had gaps to adopt more comprehensive practices that would facilitate deterrence, investigation, prosecution and enforcement
  - To set up a mechanism for timely and effective international cooperation in the area of Cybercrime

The current status of International Cybercrime treaties

- The Budapest convention had 50 signatories and it has been ratified by an additional 28 countries. It came into force on 1st July 2004.
- Apart from the 78 countries that have ratified it, 17 additional countries have observer status
- India is not a signatory to the Budapest convention. The main objections are that India did not participate in its drafting and there are concerns over sharing data with law enforcement agencies of other countries
- On 1st March 2006 an Additional Protocol was added to the convention which required ratifying states to criminalize the dissemination of racist or xenophobic material through electronic means as well as well as threats or insults motivated by racism and xenophobia
- On 8th Aug 2024 a UN committee approved the first global treaty on Cybercrime
    - This has received criticism from various quarters on grounds of expanded definitions of cybercrime, human rights and privacy

United Nations Convention against Cybercrime

- The UN General Assembly adopted resolution 79/243 on 24th Dec 2024 without vote
- The resolution among other things will open the convention for signature by member states in 2025
- The objectives as stated in the convention in Article 1 – Statement of Purpose- are

    (a) Promote and strengthen measures to prevent and combat cybercrime more efficiently and effectively;

    (b) Promote, facilitate and strengthen international cooperation in preventing and combating cybercrime; and

    (c) Promote, facilitate and support technical assistance and capacity-building to prevent and combat cybercrime, in particular for the benefit of developing countries.

Comparing the Budapest convention and the UN convention

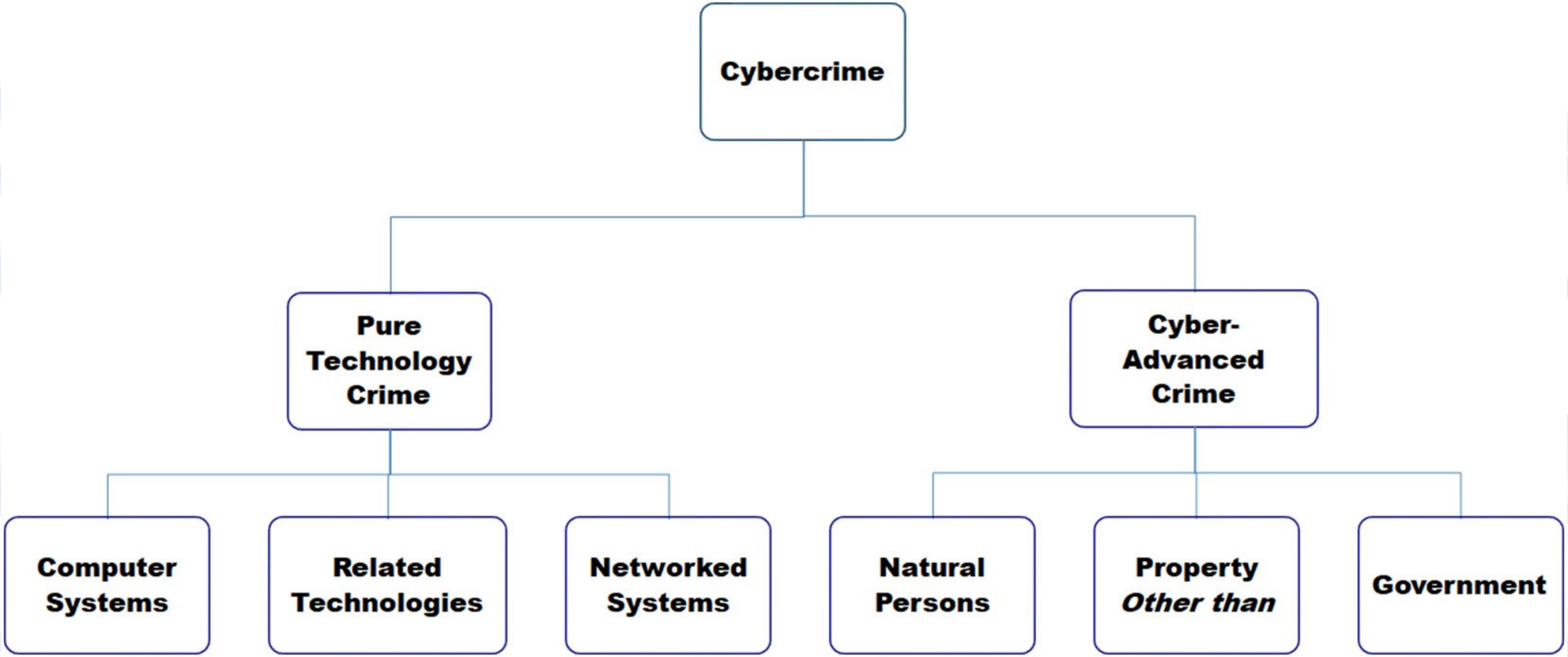
Similarities

- Both treaties aim to combat cybercrime by fostering international collaboration.
- Each provides guidelines for creating legal frameworks to address cyber offenses.
- Both emphasize capacity-building and sharing expertise to fight cybercrime.

Differences

- Inclusivity: The Budapest Convention primarily includes developed nations, while the UN Convention is designed to have broader global participation.
- Scope of Threats: The UN Convention seeks to address newer cyber threats, whereas the Budapest Convention is seen as somewhat dated in scope.
- Drafting Process: The Budapest Convention had limited global input, while the UN Convention involves broader international representation in its negotiation process.

A Taxonomy of Cybercrime



Akhilesh Chandra, Melissa J. Snowe, A taxonomy of cybercrime: Theory and design, International Journal of Accounting Information Systems, Volume 38, 2020, 100467, ISSN 1467-0895, https://doi.org/10.1016/j.accinf.2020.100467.

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Indian Cybercrime Coordination Centre (I4C)

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Home › Cybercrime categories

CYBERCRIME CATEGORIES

- Cryptocurrency Crime
- Cyber Terrorism
- Hacking / Damage to Computer Systems
- Online and Social Media Related Crime
- Online Financial Fraud
- Publishing / Transmitting of Explicit Material in Electronic Form
- Ransomware
- Child Pornography / Child Sexual Abuse Material (CSAM)

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31 OCTOBER 2024

Cybercrime

Cybercrime consists of criminal acts committed online by using electronic communications networks and information systems. The EU has implemented laws and supports operational cooperation through non-legislative actions and funding.

Cybercrime is a borderless issue that can be classified in three broad definitions:

- **crimes specific to the internet**, such as attacks against information systems or phishing (e.g. fake bank websites to solicit passwords enabling access to victims' bank accounts)
- **online fraud and forgery**: large-scale fraud can be committed online through instruments such as identity theft, phishing, spam and malicious code
- **illegal online content**, including child sexual abuse material, incitement to racial hatred, incitement to terrorist acts and glorification of violence, terrorism, racism and xenophobia

Many types of crime, including terrorism, trafficking in human beings, child sexual abuse and drugs trafficking, have moved online or are facilitated online. As a consequence, most criminal investigations have a digital component.

FBI categorization of cybercrime (Data for 2021)

By Victim Count

Crime Type	Victims	Crime Type	Victims
Phishing/Vishing/Smishing/Pharming	323,972	Government Impersonation	11,335
Non-Payment/Non-Delivery	82,478	Advanced Fee	11,034
Personal Data Breach	51,829	Overpayment	6,108
Identity Theft	51,629	Lottery/Sweepstakes/Inheritance	5,991
Extortion	39,360	IPR/Copyright and Counterfeit	4,270
Confidence Fraud/Romance	24,299	Ransomware	3,729
Tech Support	23,903	Crimes Against Children	2,167
Investment	20,561	Corporate Data Breach	1,287
BEC/EAC	19,954	Civil Matter	1,118
Spoofing	18,522	Denial of Service/TDoS	1,104
Credit Card Fraud	16,750	Computer Intrusion	979
Employment	15,253	Malware/Scareware/Virus	810
Other	12,346	Health Care Related	578
Terrorism/Threats of Violence	12,346	Re-shipping	516
Real Estate/Rental	11,578	Gambling	395

Social Engineering

Social engineering has emerged as a serious threat in virtual communities and is an effective means to attack information systems. The services used by today's knowledge workers prepare the ground for sophisticated social engineering attacks. The growing trend towards BYOD (bring your own device) policies and the use of online communication and collaboration tools in private and business environments aggravate the problem. In globally acting companies, teams are no longer geographically co-located, but staed just-in-time. The decrease in personal interaction combined with a plethora of tools used for communication (e-mail, IM, Skype, Dropbox, LinkedIn, Lync, etc.) create new attack vectors for social engineering attacks. Recent attacks on companies such as the New York Times and RSA have shown that targeted spear-phishing attacks are an eective, evolutionary step of social engineering attacks. Combined with zero-day-exploits, they become a dangerous weapon that is often used by advanced persistent threats. This paper provides a taxonomy of well-known social engineering attacks as well as a comprehensive overview of advanced social engineering attacks on the knowledge worker.

Abstract of : K. Krombholz, H. Hobel, M. Huber, and E. Weippl, “Advanced social engineering attacks,” *Journal of Information Security and Applications*, vol. 22, pp. 113–122, 2015.

What is Social Engineering and why is it a problem?

Social engineering is the art of getting users to compromise information systems. Instead of technical attacks on systems, social engineers target humans with access to information, manipulating them into divulging confidential information or even into carrying out their malicious attacks through influence and persuasion. Technical protection measures are usually ineffective against this kind of attack. In addition to that, people generally believe that they are good at detecting such attacks. Research, how- ever, indicates that people perform poorly on detecting lies and deception [42, 33]. The infamous attacks of Kevin Mitnick [35] showed how devastating sophisticated social engineering attacks are for the information security of both companies and governmental organizations. When social engineering is discussed in the information and computer security field, it is usually by way of examples and stories (such as Mitnick's). However, at a more fundamental level, important findings have been made in social psychology on the principles of persuasion. Particularly the work of Cialdini [16], an expert in the eld of persuasion, is frequently cited in contributions to social engineering research. Although Cialdini's examples focus on persuation in marketing, the fundamental principles are crucial for anyone seeking to understand how deception works.

K. Krombholz, H. Hobel, M. Huber, and E. Weippl, “Advanced social engineering attacks,” *Journal of Information Security and Applications*, vol. 22, pp. 113–122, 2015.

A Taxonomy of Social Engineering

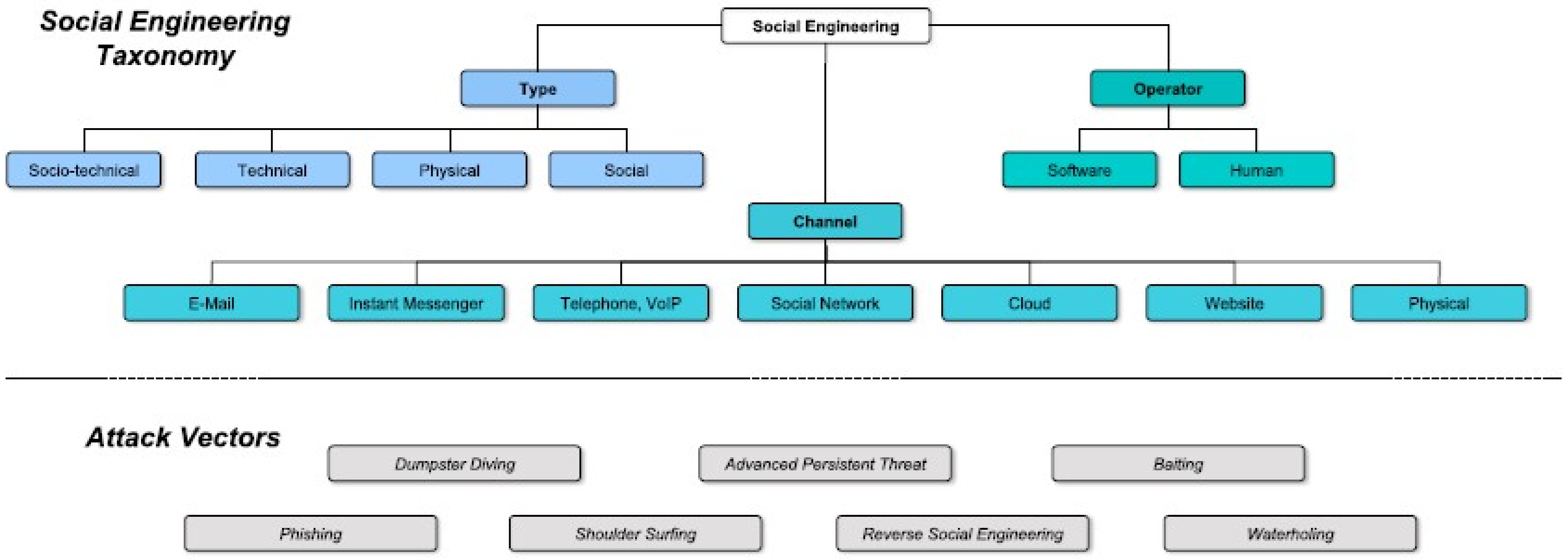


Figure 1: Overview of our classification of attack characteristics and attack scenarios.

K. Krombholz, H. Hobel, M. Huber, and E. Weippl, “Advanced social engineering attacks,” *Journal of Information Security and Applications*, vol. 22, pp. 113–122, 2015.

Global Cybercrime Trends and Incidence

https://www.economist.com/briefing/2025/02/06/online-scams-may-already-be-as-big-a-scourge-as-illegal-drugs

https://www.economist.com/leaders/2025/02/06/the-vast-and-sophisticated-global-enterprise-that-is-scam-inc

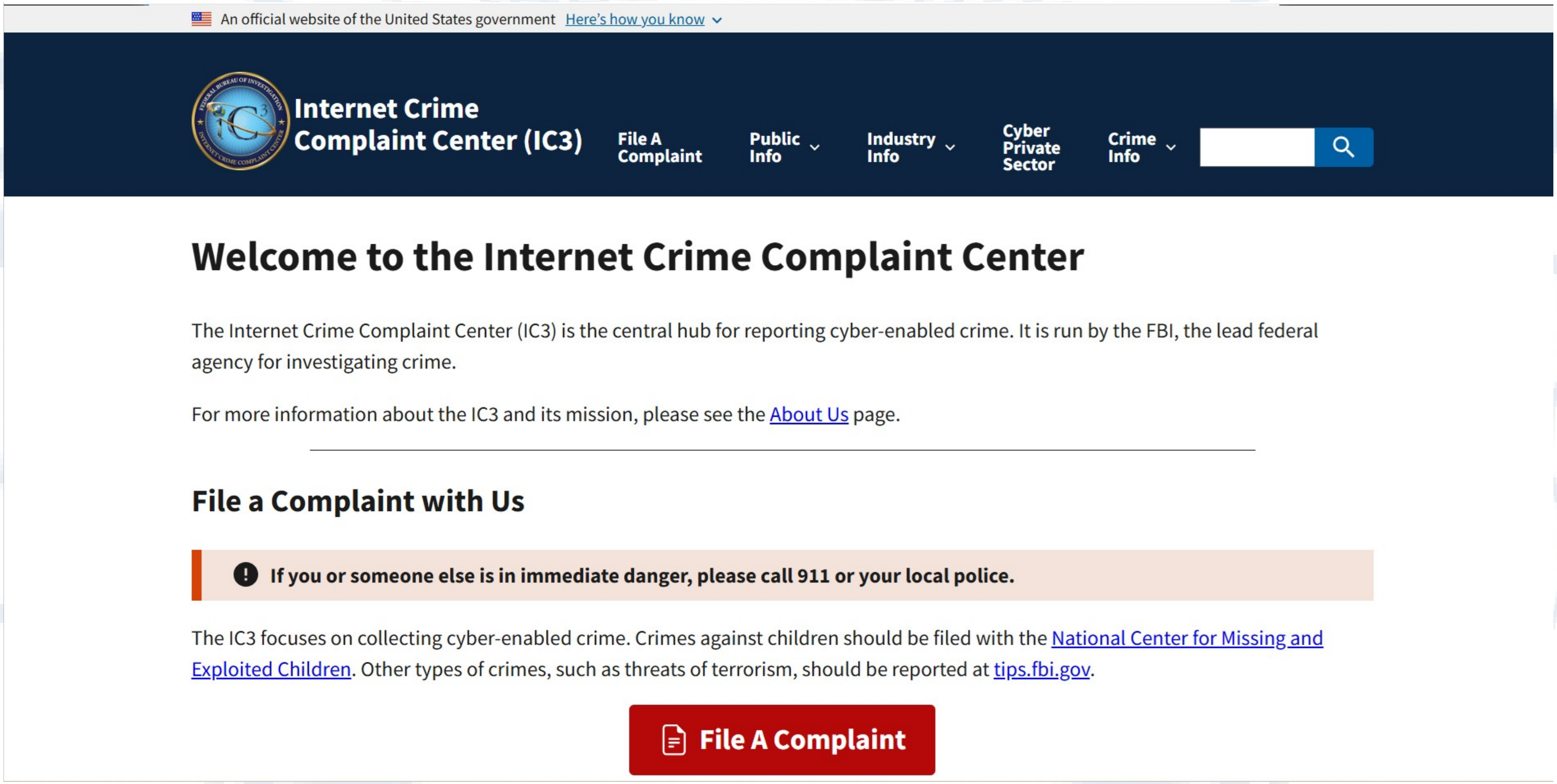
https://www.nu.edu/blog/cybersecurity-statistics/

https://indianexpress.com/article/cities/delhi/delhi-cyber-crime-cases-delhi-police-data-crypto-scam-ifso-9531556/

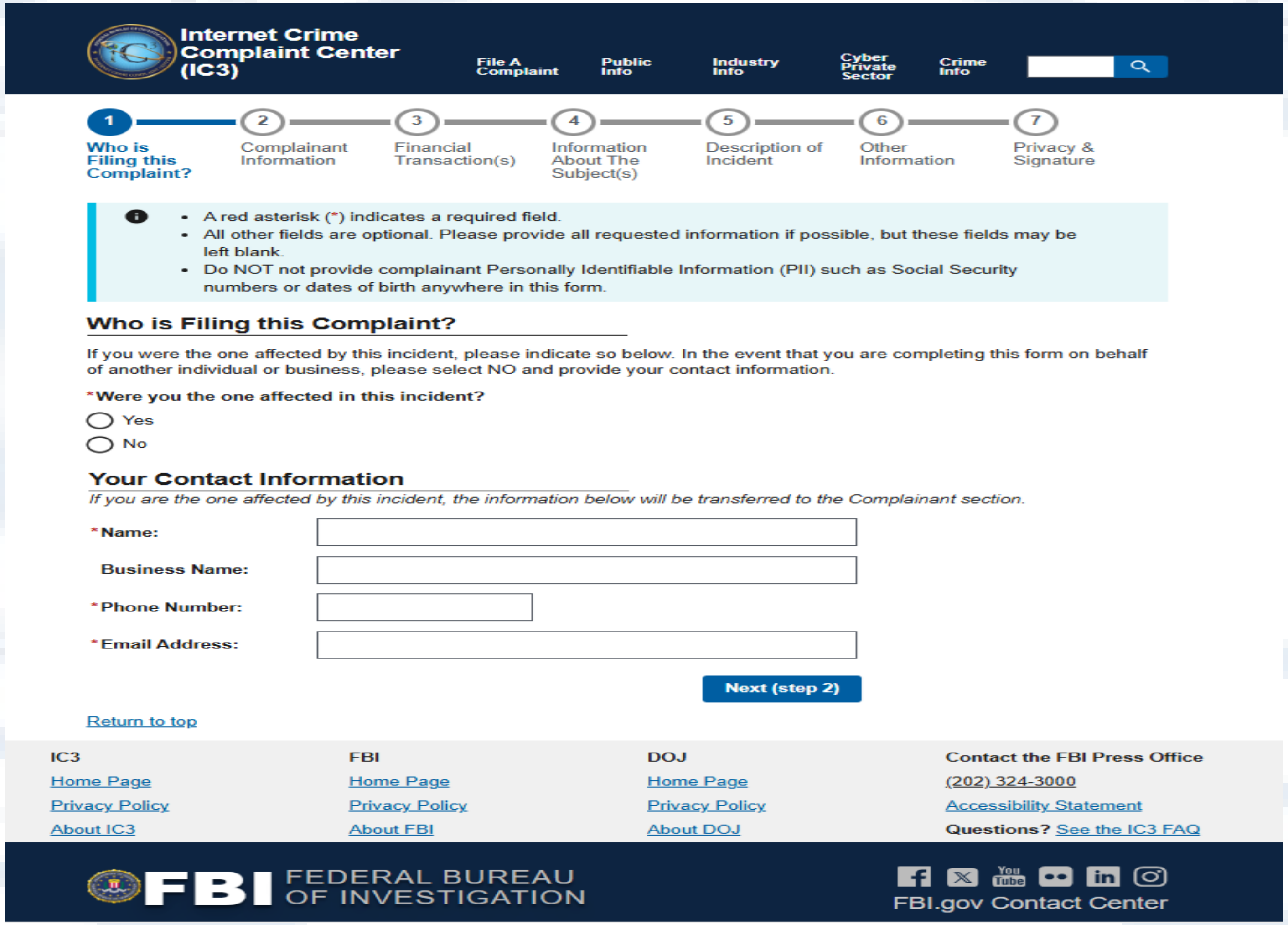
https://timesofindia.indiatimes.com/city/lucknow/massive-rs-120-crore-cyber-fraud-linked-to-aktu-two-more-arrested/a
icleshow/115310841.cms

https://www.statista.com/markets/424/topic/1065/cyber-crim
e-security/#overview

Internet Crime Complaint Centre - USA



Filing a complaint with the US IC3



Internet Crime Complaint Center (IC3)

File A Complaint | Public Info | Industry Info | Cyber Private Sector | Crime Info

1 Who is Filing this Complaint? — 2 Complainant Information — 3 Financial Transaction(s) — 4 Information About The Subject(s) — 5 Description of Incident — 6 Other Information — 7 Privacy & Signature

- A red asterisk (*) indicates a required field.
- All other fields are optional. Please provide all requested information if possible, but these fields may be left blank.
- Do NOT not provide complainant Personally Identifiable Information (PII) such as Social Security numbers or dates of birth anywhere in this form.

Who is Filing this Complaint?

If you were the one affected by this incident, please indicate so below. In the event that you are completing this form on behalf of another individual or business, please select NO and provide your contact information.

***Were you the one affected in this incident?**

- Yes
- No

Your Contact Information

If you are the one affected by this incident, the information below will be transferred to the Complainant section.

*Name:

Business Name:

*Phone Number:

*Email Address:

Next (step 2)

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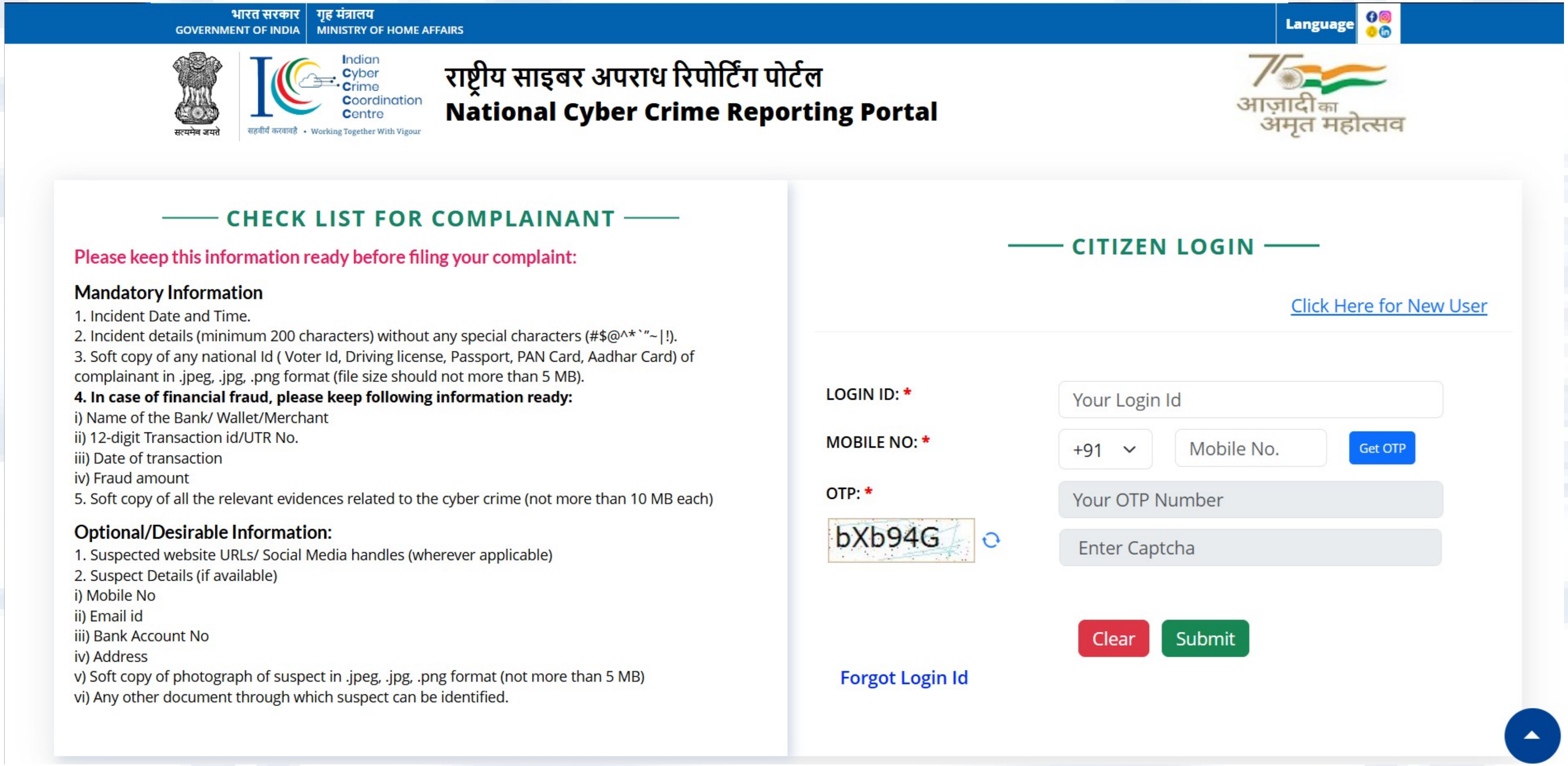
FBI.gov Contact Center

Indian Cybercrime Coordination Centre

Filing a complaint of the Indian I4C



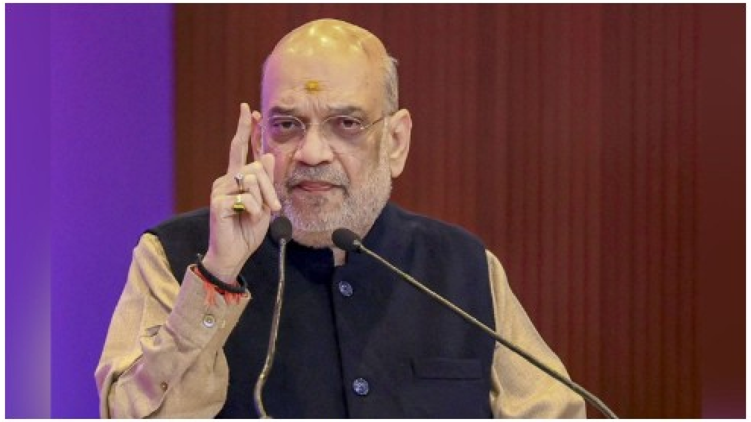
Union Home Minister Amit Shah called for coordinated and immediate action by all stakeholders to curb cyberfraud on Tuesday (February 10, 2026), stating that if steps were not taken, it would turn into a national crisis.

Stating that cybersecurity was no longer just an issue of economic security, Mr. Shah said it was now also linked to national security. “It is therefore our collective responsibility to firmly secure the economic, social, administrative, and national security dimensions, and to advance the digital revolution.”

100 victims every hour: Amit Shah warns of 'national crisis', says cybercrime has become an industry

Amit Shah said that as digital transactions have grown rapidly, the associated risks have also increased.

By: Express News Service 4 min read New Delhi Feb 11, 2026 10:51 AM IST



Amit Shah pointed out that 11 years ago, India had only 250 million internet users, whereas today it has crossed 1 billion, reaching new heights in the digital domain. (file)

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Union Home Minister Amit Shah said Tuesday that one person falls victim to cybercrime every 37 seconds, and that, on average, 100 people are affected every hour, and stressed that cybercrime has now evolved into an organised industry, where bank accounts are "bought and sold as a service."

While delivering the keynote address at a conference titled "Tackling Cyber-Enabled Frauds & Dismantling the Ecosystem," jointly organised by the CBI and the Indian Cyber Crime Coordination Centre (I4C) in New Delhi, Shah underscored the need for collective, coordinated, and immediate action against cybercrime, describing it as a growing national security threat.

https://indianexpress.com/article/india/100-victims-every-hour-amit-shah-warns-of-national-crisis-says-cybercrime-has-become-an-industry-10525831/

SC to hear suo motu case on ‘digital arrest’ cyber fraud next week

Yekkirala Akshitha March 17, 2026

The Supreme Court of India on Monday said it would hear next week a **suo motu case on victims of “digital arrest” scams** , a fast-growing cybercrime in which fraudsters impersonate law enforcement or government officials to extort money.

The matter was mentioned by Attorney General R Venkataramani before a bench headed by Chief Justice Surya Kant and Justice Joymalya Bagchi. The top law officer informed the court that a status report on measures taken to tackle the fraud would be placed before it during the day and requested that the hearing be taken up next week. The bench agreed and said the case would be listed again at the earliest.

Digital arrest scams have emerged as a major cybercrime in India. In such frauds, criminals pose as officials from agencies such as the police, the Central Bureau of Investigation, or telecom and financial regulators. Through threatening video or audio calls, they falsely claim the victim is under investigation or “digitally arrested” and pressure them to transfer money to so-called verification or safe accounts. Victims are often kept on prolonged calls and psychologically coerced into making payments.

On February 9, the apex court took serious note of the scale of such cyber frauds, observing that more than **Rs 54,000 crore had been siphoned off** , an amount it described as nothing short of “robbery or dacoity”. The court said the losses were larger than the annual budgets of several small states and called for urgent coordinated action.

The bench directed the Centre to prepare a **standard operating procedure (SoP)** to deal with digital arrest scams in consultation with key stakeholders including the Reserve Bank of India, banks and the Department of Telecommunications.

https://tmv.in/article/sc-to-hear-suo-motu-case-on-digital-arrest-cyber-fraud-next-week-date=2026-03-17

To summarize

- The vulnerabilities of software, the complications of working with many jurisdictions and the ease with which people can be deceived online create an environment where cybercrime is rampant
- Different states have different laws and classifications for cybercrime
- States vary greatly in their ability to investigate cybercrime and successfully prosecute cybercriminals
- Investigation and prosecution is greatly complicated by the fact that most cyberattacks intentionally involve many jurisdictions
- There is ongoing effort to increase international cooperation in the area of deterring and bringing cybercriminals to book. The UN convention on cybercrime is one of the latest of these efforts
- Sadly however at the present time the incidence and impact of cybercrime seems to be increasing

End of

Lecture #15